

**UNIVERSITY OF SINDH, JAMSHORO**

**Department of Artificial Intelligence**

**Assignment No. 1**

**Course: Introduction to Artificial Intelligence**

**Topic: PEAS Framework and Task Environment  
Analysis**

**Selected AI Application: AI Tutor for Personalized  
Student Learning Recommendations**

**Submitted By:**

**Name: Ali Muhammad**

**Roll Number: 2K24/AIE/07**

**Department: Artificial Intelligence**

**Submitted To:**

**Sir Rajesh**

**University of Sindh, Jamshoro**

## **TASK 1: PEAS SPECIFICATION**

### **Introduction**

Artificial Intelligence is increasingly being used in education to improve learning experiences. An AI Tutor is an intelligent system that analyzes student performance and provides personalized learning recommendations. It helps students learn according to their abilities and progress.

### **1. Performance Measures**

- Learning Improvement Rate – Measures how much a student's knowledge improves over time.
- Recommendation Accuracy – Determines whether the suggested learning materials are relevant and useful.
- Course Completion Rate – Measures the percentage of students who successfully complete their courses.
- Student Satisfaction Score – Evaluates how satisfied students are with the learning experience.
- Assessment Performance – Tracks student results in quizzes, assignments, and examinations.
- Engagement Level – Measures how actively students participate in learning activities.

### **2. Environment**

The AI Tutor operates in an online educational environment that includes students with different learning abilities, learning management systems, educational resources, teachers, and internet-connected devices.

### **3. Actuators**

Personalized learning recommendations, quiz suggestions, automated feedback generation, study schedule recommendations, notifications, reminders, and adaptive content delivery.

### **4. Sensors**

Quiz scores, examination results, assignment performance, student learning history, time spent on educational content, and student feedback.

## **TASK 2: ENVIRONMENT CLASSIFICATION**

### **Partially Observable**

The AI Tutor cannot directly observe a student's actual understanding of a topic and must rely on performance data.

### **Single-Agent**

The AI Tutor acts as the primary decision-making system that generates recommendations independently.

### **Stochastic**

Student behavior is unpredictable and outcomes cannot always be determined with certainty.

### **Sequential**

Current recommendations influence future learning outcomes and decisions.

### **Dynamic**

Student knowledge, interests, and performance continuously change over time.

### **Continuous**

Learning progress and performance change gradually rather than in fixed states.

## **TASK 3: CRITICAL ANALYSIS**

### **Question 1**

The greatest challenge is the Partially Observable environment. An AI Tutor cannot directly determine whether a student truly understands a concept. Because the system receives incomplete information, it must make intelligent predictions about student knowledge. Developing accurate models for such predictions is a significant challenge for AI engineers.

### **Question 2**

A suitable utility function is:

$$U = 0.7(\text{Learning Accuracy}) - 0.3(\text{Time Cost})$$

This utility function encourages the AI Tutor to maximize learning effectiveness while minimizing unnecessary study time. If the weight of Learning Accuracy is doubled, the AI Tutor will focus more on improving educational outcomes and may recommend additional exercises and practice sessions.

#### TASK 4: STRUCTURED REPRESENTATION (JSON)

```
{
  "system_name": "AI Tutor for Personalized Student Learning Recommendations",
  "peas": {
    "performance_measures":
    ["learning_improvement_rate", "recommendation_accuracy", "course_completion_rate", "student_satisfaction_score"],
    "environment": "Online educational platform",
    "actuators": ["personalized_recommendations", "quiz_suggestions", "automated_feedback"],
    "sensors": ["quiz_scores", "exam_results", "learning_history"]
  },
  "environment_classification": {
    "observability": {"choice": "Partially Observable"},
    "agents": {"choice": "Single-Agent"},
    "determinism": {"choice": "Stochastic"},
    "episodic_sequential": {"choice": "Sequential"},
    "static_dynamic": {"choice": "Dynamic"},
    "discrete_continuous": {"choice": "Continuous"}
  },
  "utility_function": "U = 0.7(Learning Accuracy) - 0.3(Time Cost)"
}
```

#### Conclusion

The AI Tutor is an effective educational application of Artificial Intelligence that provides personalized learning support to students. Through continuous monitoring and intelligent recommendations, it improves learning outcomes and enhances the educational experience.